One coupling interval, repeated for the whole run

Gas chemistry and aerosol take turns (operator splitting). In each interval Δt_c (5 min in every level, 1 min with nucleation on) the chemistry runs first, its products go to the aerosol model, and the vapors the particles took up or released go back to the gas. PPM marks where particles move between size bins with the piecewise-parabolic method.

Scenario

- initial gases: limonene, O_3, NO, NO_2, SO_2, CO, CH_4 …
- seed aerosol: number, size, sulfate / organic (goes straight to step 3)
- T, RH (constant for the run); site and date (Level 3)
- $\gamma(N_2O_5)$; c* table (Li et al. 2016 for species not in it); number of size bins (40–320)

1 Gas-phase chemistry

Photolysis rates J
Level 3: TUV-x from latitude, longitude, date and time (at the middle of Δt_c).
Level 2: constant light (a 30° sun scaled to $J(NO_2) = 5\times10^{-3}\ s^{-1}$, about half).

Rate constants for all 2135 reactions, frozen for Δt_c: MCM expressions, J, the RO_2 sum, and $k_{het}(N_2O_5)$ from the particles.

Stiff ODE solver JAX
MCM v3.3.1 limonene subset, 725 species (diffrax Kvaerno3, rtol 10^{-3}), integrated over Δt_c.

2 Hand-off

Sulfuric acid
$SO_2 + OH \rightarrow H_2SO_4$, counted over Δt_c → a production rate

Nitrate
N_2O_5 taken up by particles → 2 NO_3^- per N_2O_5 → a production rate

Condensable organics
228 products grouped by c* into 15 volatility classes (half decades, 10^{-3} to 10^4 µg m^{-3}) — the VBS gas phase. c* from S. Jathar's table (Nannoolal et al. 2008, via PyCHAM)

3 Aerosol microphysics

TOMAS sectional model, 60 s micro-steps over Δt_c (5 s with nucleation; page 2)

- sources: H_2SO_4 vapor; nitrate onto particles PPM
- nucleation (H_2SO_4–H_2O; off in the presets)
- coagulation
- H_2SO_4 condensation PPM
- **SOA partitioning** (vbs-jax): condensation and evaporation per class and size bin PPM
- water uptake (equilibrium)

4 Hand-back

- change of each class's vapor → back to the individual species (by gas share when condensing, by particle share when evaporating)
- per-species particle mass → "top contributors"
- Level 3: NO_2 reset to 1 ppb (stand-in for emissions)

particle surface area → $k_{het}(N_2O_5)$ and each bin's nitrate share, for the next interval

next interval: gases changed by condensation and evaporation (vapors can react again)

What the particles carry

- **sulfate**: the seed plus condensed H_2SO_4
- **POA**: the seed's non-volatile organic; absorbs SOA
- **SOA**: 15 volatility classes; a per-species tracker shows which products they hold
- **nitrate**: only from N_2O_5 uptake (non-volatile)
- **water**: in equilibrium with RH

Not in the model: ammonium nitrate and NH_3, HNO_3 partitioning, sea salt, dust; dilution, deposition and wall loss.

The three levels

- **Level 1, Volatility**: no chemistry (steps 1, 2 and 4 are skipped). A surrogate organic with the c* you choose is produced straight into one class; H_2SO_4 at a production rate you choose.
- **Level 2, Limonene SOA**: full chemistry, constant light, 6 h.
- **Level 3, Day and night**: TUV-x light over 24 h, N_2O_5 uptake (γ) makes nitrate at night, NO_2 reset to 1 ppb every interval Δt_c (5 min; 1 min with nucleation on).

Clocks and grids

- coupling interval Δt_c, a numerical solver setting: 300 s in every level (Level 3 used 900 s until 17 Oct 2026). A result X has a splitting error that grows with it, $X \approx X_0 + a\cdot\Delta t_c$ (X_0: X without splitting). Planned: the solver picks Δt_c.
- aerosol micro-step Δt_m: 60 s; vbs-jax sub-steps inside it
- with nucleation on: Δt_m = 5 s, and the requested interval Δt_{req} is cut into $m = \lceil \Delta t_{req} / 60\ s \rceil$ equal parts, $\Delta t_c = \Delta t_{req} / m$ (60 s for 300, 600 or 900 s); output times unchanged
- size bins: 40 (default), 80, 160 or 320, always 1.7 nm to 17.5 µm at sulfate density (plots: 1.8 nm to 19 µm)
- output: every 10 min (Levels 1–2), 30 min (Level 3)
- all in float64 JAX; one compile per grid and level

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Inside one aerosol micro-step (60 s; 5 s with nucleation on)

Step 3 on page 1, in the order the model runs it. Each step changes the size distribution (number and mass of each species in each size bin) before the next one starts.

1 **Sources.** The H_2SO_4 vapor C gets the micro-step's production at once, $C \leftarrow C + P \cdot \Delta t_m$ (P the rate from the chemistry, Δt_m the micro-step). Nitrate from N_2O_5 uptake is added to the particles, each size bin getting its share of the uptake (larger surface, larger share); the particles that grew are then moved to the bins that match their new mass PPM.
Then MNFIX MNFIX keeps each bin's mean mass inside the bin. Level 1: the surrogate vapor and H_2SO_4 come from your production rates.

2 **Nucleation.** New 1.7 nm particles from H_2SO_4 and water at the rate $J = k(T) \cdot C^n$, n ≈ 4, k(T) larger in colder air (the binary neutral term of Dunne et al. 2016; NH_3, ions and organics set to zero); as in TOMAS each new particle is 10 % organic by mass. Off in the presets; it needs high H_2SO_4 and cold air.
Why 5 s: C jumps by $P \cdot \Delta t_m$ at each micro-step's start and J grows as C^4, so 60 s steps put the peak particle number up to 78 % too high; 5 s is within 4 % (40 bins). Sub-steps Δt_s keep $J \cdot \Delta t_s \le N/2$ (N the total particle number); MNFIX afterwards MNFIX.

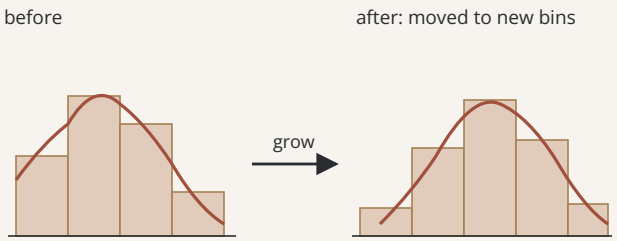
3 **Coagulation.** Particles collide and stick: number falls, mass stays. 40 bins: the two-moment scheme of Fortran TOMAS (Tzivion, Feingold and Levin 1987); 80–320 bins: the linear discrete method (Simmel et al. 2002).
Ten forward-Euler sub-steps $x \leftarrow x + (\Delta t_m/10) \cdot dx/dt$, x each bin's number and mass (6 s each; 0.5 s with nucleation on), with the collision kernel fixed over Δt_m; MNFIX after each MNFIX.

4 **H_2SO_4 condensation.** The vapor condenses onto every bin at the kinetic rate (Fuchs–Sutugin correction in Dahneke's form, accommodation 1); it never evaporates. Particles grow across bin edges PPM.
Flux-form PPM in log-mass with a monotone limiter (Colella and Woodward 1984); MNFIX before and after MNFIX, then particle water is reset.

5 **SOA partitioning (vbs-jax).** For each volatility class and size bin: condensation where the vapor exceeds the bin's equilibrium vapor, evaporation where it is below. Equilibrium vapor = c* (corrected to the run temperature) × the class's share of the organic mass (Raoult, by mass) × the Kelvin term (σ = 0.025 N m^{-1}). Seed POA and SOA absorb the vapor; a particle with no organic yet still takes up vapor above its c* (absorbing floor, ASSUMPTIONS AC1). Then each bin's particles move by their own growth or shrinkage PPM.
vbs-jax adaptive sub-steps with a Raoult fixed-point loop; the PPM remap carries each bin's parabolic sub-bin profile and its composition to the bins that match the new masses; organic that would push the top bin off the grid stays in the gas; MNFIX keeps a tiny floor in empty bins MNFIX.

6 **Water.** Particle water in equilibrium with RH: ammonium-bisulfate water ratio for sulfate and, as a simplification, for organics and nitrate too (organics come out too hygroscopic; ASSUMPTIONS A9).

What PPM does here



A sectional model keeps particles in fixed size bins. When particles grow, they must be moved to bigger bins without smearing the distribution or losing mass. PPM fits a parabola to the number in each bin (limited, so it never goes negative or overshoots), shifts it by the growth, and deposits it onto the fixed bins. Number and the mass of every species are conserved to round-off.

- **Used for**: H_2SO_4 condensation, SOA growth and evaporation, nitrate added by N_2O_5 uptake.
- **Before PPM** (tag aaar-stable-2026-10-15): SOA and nitrate were moved by MNFIX, which made the size distribution jagged; with PPM the mean diameter changes by under 0.6 % from 40 to 160 bins.

MNFIX, still used

TOMAS's bookkeeping fix (Adams and Seinfeld 2002): if a bin's mean particle mass has left the bin (after steps 1–5), part of the bin moves to the neighbouring bin so number and mass stay consistent. With PPM it only does small corrections; the old SOA and nitrate path (moved by MNFIX) is kept for comparison.

References

Adams, P. J. and Seinfeld, J. H.: Predicting global aerosol size distributions in general circulation models, J. Geophys. Res., 107(D19), 4370, 2002. · Colella, P. and Woodward, P. R.: The Piecewise Parabolic Method (PPM) for gas-dynamical simulations, J. Comput. Phys., 54, 174–201, 1984. · Tzivion, S., Feingold, G. and Levin, Z.: An efficient numerical solution to the stochastic collection equation, J. Atmos. Sci., 44, 3139–3149, 1987. · Simmel, M., Trautmann, T. and Tetzlaff, G.: Numerical solution of the stochastic collection equation – comparison of the Linear Discrete Method with other methods, Atmos. Res., 61, 135–148, 2002. · Zaveri, R. A. et al.: Modeling kinetic partitioning of secondary organic aerosol and size distribution dynamics, Atmos. Chem. Phys., 14, 5153–5181, 2014. PDF · Pankow, J. F.: An absorption model of gas/particle partitioning of organic compounds in the atmosphere, Atmos. Environ., 28, 185–188, 1994. · Li, Y., Pöschl, U. and Shiraiwa, M.: Molecular corridors and parameterizations of volatility, Atmos. Chem. Phys., 16, 3327–3344, 2016. PDF · Nannoolal, Y., Rarey, J. and Ramjugernath, D.: Estimation of pure component properties. Part 3. Estimation of the vapor pressure of non-electrolyte organic compounds via group contributions and group interactions, Fluid Phase Equilib., 269, 117–133, 2008.

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